



Script 2 — Disinfection Byproducts, Lead and Copper, and Public Communication

Recording script. Target ~15–20 minutes at a normal reading pace. Headers, bracketed notes, and timing estimates are for your reference and are not read aloud.

1. The Disinfection Tradeoff [~1 min]

The last script ended on disinfection as a barrier against pathogens. Disinfection solves one problem but creates another. When disinfectants react with the natural organic matter in the water, they form disinfection byproducts which are linked to cancer over a lifetime of exposure. A system is always balancing two opposing pressures. The Surface Water Treatment Rules push disinfectant dose and contact time up to kill pathogens while the disinfection byproduct rules push the byproducts down, and cap how much disinfectant residual is carried. The CT you need for inactivation also forms byproducts, so you cannot simply dose your way out of either problem. Managing that balance is a core concept of this script.

2. The Stage 1 and Stage 2 Disinfection Byproducts Rules [~7 min]

The disinfection byproduct rules consist of two rules that target reducing the byproducts formed when disinfectants react with organic matter.

Stage 1 set the MCLs for the byproducts. Total trihalomethanes, or TTHM at 80 parts per billion, haloacetic acids, or HAA5 at 60 parts per billion, bromate at 10 parts per billion, and chlorite at 1.0 milligram per liter.

Compliance for TTHM and HAA5 was determined on a system-wide running annual average. Sample the designated locations quarterly, average the last four quarters, and compare that average to the MCL. Each new quarter, the oldest quarter drops off.

Stage 1 also set the maximum residual disinfectant limits, or MRDLs, which are ceilings on the disinfectant itself. Chlorine and chloramines at 4.0 milligrams per liter and Chlorine dioxide at 0.8 milligrams per liter.

The rule lets an operator push chlorine or chloramine residual above the MRDL temporarily to deal with specific microbial problems, such as those occurring due to a main break or a contamination event, so long as the State is notified. Chlorine dioxide is the exception; its MRDL cannot be exceeded.

Stage 1 also set precursor-removal requirements through enhanced coagulation. Surface water systems using conventional filtration must remove a set percentage of total organic carbon, or TOC. The required percentage is derived by comparing the source-water TOC against its alkalinity. Byproducts form when disinfectant meets organic matter, so removing organic matter before disinfection cuts byproduct formation at the source.

Stage 2 kept the same MCLs and the same MRDLs, but changed how compliance is measured to close an underlying gap identified in Stage 1. Under the system-wide running annual average, a system with several good sampling sites and one bad one could still achieve MCL compliance by diluting the average. Through this flaw, a system could maintain compliance while serving non-compliant water to the portion of consumers connected to the bad site. Stage 2 fixed this by replacing the system-wide running annual average with a locational running annual average, or LRAA. Now each sampling location is averaged on its own and must meet the MCL independently. A hot spot can no longer hide inside the system-wide number.

Stage 2 also added the operational evaluation level, or OEL, which is an early-warning trigger, not a violation. It is calculated as the two previous quarters, plus twice the most recent quarter, all divided by four. Doubling the most recent quarter makes a rising trend show up fast. If the OEL comes out higher than the MCL, the system must perform an operational evaluation of what is driving byproduct formation and report to the State within 90 days. This approach forces action before a rising trend becomes an actual violation and threatens public health.

Stage 2 also required an initial distribution system evaluation, or IDSE. To identify Stage 2 sampling sites, a system had to find the locations within its distribution system with the highest byproduct levels. Those sites became its monitoring locations so that compliance is judged at the worst locations, not the best.

With both rules in view, controlling byproducts day to day comes down to a handful of approaches.

First, take the precursors out before you disinfect. Beyond enhanced coagulation and enhanced softening, systems can use granular activated carbon, or membrane filtration to reduce TOC concentration. Less organic matter going in means less byproduct coming out.

Second, manage the disinfectant. Systems can optimize the chlorine dose so it is no higher than it needs to be, move the point of chlorination downstream to reduce contact time with organics, or lean on non-chlorine forms of disinfection, like ozone, to achieve disinfection credit. The tradeoff is that each alternate disinfectant creates disinfection byproducts of its own. A system needs to manage its disinfectant and CT choices with respect to the concentration and types of byproducts formed. In addition to managing disinfectants at the treatment plant, systems can also switch the distribution residual from free chlorine to chloramines, which forms far less TTHM and HAA5, but introduces the possibility for nitrification and a byproduct of its own in the form of N-nitrosodimethylamine, or NDMA. NDMA has no MCL in California today, but the State Water Board has a rulemaking in progress to set one. What these approaches have in common is that they lower the byproducts leaving the plant, but none of them stops the reaction. A system must carry a disinfectant residual into the distribution system, where byproducts continue to form as the water travels and ages.

Third, reduce the water age. This is the reason managing the disinfectant is only half the job. TTHM and HAA5 keep forming the entire time chlorinated water sits in the system, so a plant can dose perfectly

and still build byproducts out in the distribution system where the water is oldest. Turning tanks over, flushing, and looping dead ends all pull those levels down where they accumulate.

One additional approach works on only one byproduct. TTHMs are volatile and can be removed via aeration or air stripping. Spraying water or adding aerators or air stripping towers at strategic locations can reduce TTHM levels. HAA5 is not volatile, so aeration does nothing for it.

Lastly, Stage 2 and LT2 were finalized together. There is a real risk that a system chasing byproduct compliance will underdose its disinfectant and let pathogen control slip, so the two rules were built to protect both sides at once. LT2 guards pathogen protection by leaning on disinfection profiling and benchmarking. Disinfection profiling and benchmarking was introduced in the Interim Enhanced Surface Water Treatment Rule and LT1 and required systems build a disinfection profile which is a record of the microbial inactivation it achieves, and from it establish a benchmark which is its lowest monthly inactivation. What matters with LT2 is the “significant change” requirement. Before a system makes a significant change to its disinfection practices, meaning the type of disinfectant, the point where it is added, or the disinfection process, it must consult the State and use its benchmark to weigh whether the change would weaken microbial protection. The point is to keep a system from cutting disinfection just to lower its byproducts.

3. The Lead and Copper Rule [~5 min]

[Volatile section. Numbers and dates below are as of August 2026. Re-verify before the exam.]

The Lead and Copper Rule protects public health by minimizing lead and copper at the tap. Lead and copper almost never come from the source water. When water is corrosive, lead and copper can leach in from the plumbing via lead service lines, lead solder, brass fixtures, and copper pipe. Because the problem is the plumbing and not the source, this is a treatment-technique rule, not an MCL rule. There is no safe level of lead, especially for young children and pregnant women.

The rule currently sits between regulatory updates, both approved and pending final implementation. In 2021, the Lead and Copper Rule Revisions, or LCRR, required systems to inventory their service lines and catalog the types within, provide that inventory to the State by October 16, 2024, and send annual notices to those customers served by lead, galvanized requiring replacement, or unknown determination. The LCRR also required a lead action level, or AL, exceedance to be followed up by a Tier 1 public notification. The numbers in force now are an action level of 15 parts per billion for lead, and 1.3 milligrams per liter for copper, both of which are measured at the 90th percentile of first-draw samples from high-risk taps.

Exceeding an action level is not itself a violation. It is a trigger. Once exceeded, several responses are required. Although exceeding either the lead or copper action level triggers a response, the rules evolution has centered increasingly on lead as the primary contaminant of concern, and the responses required today highlight that.

First, public education. Because corrosion happens in the home's own plumbing, a system cannot fix every tap on its own, so the rule requires notifying customers within 60 days of an exceedance with information on what they can do to limit lead exposure within their home. Examples include installing point of use treatment systems, such as an under the sink filtration unit, running their tap after water has sat overnight before drinking, not cooking with hot tap water since hot water leaches more lead out of the plumbing, and getting their water tested if they're worried. This step is triggered specifically by a lead exceedance, not copper. Notification must be repeated at least annually, and a brief message included on water bills at least quarterly.

Second, corrosion control treatment, or CCT. CCT alters the water chemistry so that it stops dissolving lead and copper from the plumbing in the first place. There are three ways to do it. First, raising the water's pH with a chemical like caustic soda or lime renders the water less corrosive and promotes the creation of a protective coating on the inside of pipes. It's usually the cheapest option, but raising pH can work against your disinfection and change which disinfection byproducts form, so monitoring the two practices in tandem is crucial. The second CCT option is dosing a corrosion inhibitor. Feeding a chemical like orthophosphate creates a protective film on pipe walls. Carrying a residual all the way to the tap keeps that film in place. The tradeoff is that the phosphate adds load to the downstream wastewater plant and requires a steady, reliable feed. The final approach is calcium hardness adjustment, laying down a thin calcium carbonate film on the pipe, although this treatment approach is mainly for soft, low-calcium water and is rarely the primary fix. In any case, a system studies which of these options is the best treatment candidate and the State designates which one you use.

Alongside corrosion control treatment, the system monitors water quality parameters like pH and alkalinity to prove the corrosion control is working and does source-water monitoring to confirm that the lead and copper are coming from the plumbing and not the source. In the unlikely event that the source water is contributing, source-water specific treatment is added.

Under the rules in force now, if the lead AL is still exceeded after corrosion control treatment and any source water treatment, removing lead service lines becomes mandatory. However, on the horizon is the Lead and Copper Rule Improvements, or LCRI. As of mid-2026, the LCRI is final, with an implementation date of November 1, 2027. On that date, the rule tightens its stance on lead by decreasing the lead AL from 15 to 10 parts per billion and requires systems to replace all their lead and galvanized requiring replacement service lines, regardless of sampling results, within roughly 10 years.

Although the LCRI is final it is under legal challenge, and a court could change the timeline, so treat the 10 parts per billion number and the full-replacement mandate as final but contested.

4. The Public Notification Rule [~5 min]

The Public Notification Rule sets how fast, and by what means, a system must notify the public when something is wrong with their water. It is built on three tiers, ranked by how urgent the risk is.

Tier 1 is the most urgent and requires immediate consultation with the State and public notice within 24 hours. Tier 1 covers water quality concerns that are acute in nature that have the potential to make people sick right away such as:

- *E. coli* in the distribution system.
- Exceedance of the CFE maximum turbidity level if elevated to Tier 1 by the State.
- A nitrate or nitrite MCL violation.
- A chlorine dioxide MRDL violation in the distribution system.
- A lead action level exceedance.

Tier 2 notifications are required within 30 days and repeated every 3 months until the problem is resolved. These risks matter, but the consequences are chronic in nature, building over time rather than hitting immediately and include:

- All MCL, MRDL, and treatment-technique violations except where Tier 1 is required.
- Arsenic MCL exceedance.
- Disinfection Byproduct MCL exceedance.
- Failure to maintain filtration or disinfection requirements.
- Failure to maintain 4-log treatment for more than 4 hours under the Ground Water Rule.

Tier 3 notifications are required within 12 months and repeated annually while the issue is unresolved. These cover monitoring and testing-procedure violations, sampling and paperwork failures, or scenarios where you cannot confirm the water's quality but there is no known immediate problem. A Tier 3 notice is typically delivered as part of the annual Consumer Confidence Report.

Every Public Notification must include 10 pieces of information:

- what happened and which contaminant;
- when it happened;
- the potential health effects, in the standard required language;
- who is at risk;
- whether to use an alternate water supply;
- what actions to take;
- what the system is doing to fix it;
- when it expects to be back in compliance;
- a contact for more information;
- and language urging people to share the notice with others who drink the water.

Delivery must reach consumers. Tier 1 goes out by fast, broad methods such as radio, television, public posting, or hand delivery. Tier 2 and Tier 3 usually go by mail or direct delivery, with additional

methods to catch anyone the first method missed. Where a large share of customers do not speak English, the notice must reach them in the appropriate language.

5. The Consumer Confidence Report Rule [~3 min]

The Consumer Confidence Report, or CCR, is the annual water-quality report that every community water system sends its customers. It is the routine, once-a-year transparency document, as opposed to public notification, which is the alarm for when something goes wrong.

The report goes out to customers and to the State each year by July 1, covering the previous calendar year from January through December. A system serving 100,000 or more people must also post it on a public website, and every system must provide a copy on request.

The report must contain 8 pieces of information, in plain terms:

- the system's name and a contact;
- the sources of the water;
- definitions of terms like MCL and action level;
- a table of the contaminants detected, with their levels, the matching MCLs, and the likely source of each;
- information on any required monitoring for things like *Cryptosporidium*;
- a statement of compliance, including any violations during the year;
- variances and exemptions, if applicable;
- and required educational language, including specific statements about nitrate, arsenic, and lead, and information for vulnerable people about *Cryptosporidium*.

Under the recent CCR revisions, starting in 2027 a system serving 10,000 or more people must issue the report twice a year instead of once. That same year, the website-posting requirement expands to reach more systems, dropping from 100,000 people down to 50,000. The revisions also allow electronic delivery by emailing the report or a link to it.

6. Close [~1 min]

The through-line of this script is balance and communication. The disinfection byproduct rules balance pathogen control against the byproducts of disinfection, using enhanced coagulation to cut precursors, location-based averaging to catch hot spots, and disinfection profiling to protect pathogen control when practices change. The Lead and Copper Rule treats corrosion as the real problem and is shifting from managing lead to removing it. The Public Notification Rule and the Consumer Confidence Report are the two ways a system tells the public the truth about its water: public notification when something is urgent, and the confidence report once a year as a matter of routine.